

YallaExpand Intern Tasks - Day 1

Welcome! Please complete the following tasks and submit your solution.

Submission Instructions

Google Drive Folder:

<https://drive.google.com/drive/folders/1N-p-yXn3FISrGbnIIrtmw170U4QbEFFx?usp=sharing>

Before uploading:

- Create a folder named with your full name.
- Inside your folder, create another folder named 'Day1 Tasks'.
- Upload all solutions for today's tasks inside the 'Day1 Tasks' folder.
- You may submit separate files or a single project containing all tasks.

Task 1: Implement curry(fn)

Objective:

Implement a curry function that transforms a function of N arguments into a sequence of function calls until all required arguments are provided, then invokes the original function.

Requirements:

- Support partial application.
- Accept one or more arguments per call.
- Invoke the original function only after receiving all required arguments.

Example:

```
function curry(fn) {  
  // your implementation  
}
```

```
function sum(a, b, c) {  
  return a + b + c;  
}
```

```
const curriedSum = curry(sum);
```

```
curriedSum(1)(2)(3);    // 6  
curriedSum(1, 2)(3);    // 6  
curriedSum(1)(2, 3);    // 6  
curriedSum(1, 2, 3);    // 6
```

Task 2: Implement myPromiseAll

Objective:

Implement your own version of Promise.all.

Requirements:

- Resolve with results in the same order as the input.
- Reject immediately if any promise rejects.
- Treat non-promise values as already resolved.
- Resolve with [] when given an empty array.

Example:

```
function myPromiseAll(promises) {  
  // your implementation  
}
```

```
myPromiseAll([  
  Promise.resolve(1),  
  2,  
  new Promise((res) => setTimeout(() => res(3), 100)),  
]).then(console.log); // [1, 2, 3]
```

Task 3: Implement Array.prototype.myMap

Objective:

Implement Array.prototype.myMap matching native map behavior.

Requirements:

- Callback signature: (element, index, array)
- Support optional thisArg.
- Return a new array without mutating the original.
- Correctly skip holes in sparse arrays.
- Work with any array and callback.

Example:

```
[1, 2, 3].myMap((x, i, arr) => x * 2);  
// [2, 4, 6]
```

```
[1, 2, 3].myMap(function(x) {  
  return x + this.offset;  
}, { offset: 10 });  
// [11, 12, 13]
```

```
[1, , 3].myMap(x => x * 2);  
// [2, <1 empty item>, 6]
```

Submission Checklist

- Create your folder using your full name.
- Create a 'Day1 Tasks' folder inside it.
- Upload all required solutions.
- Ensure your code runs correctly before submission.